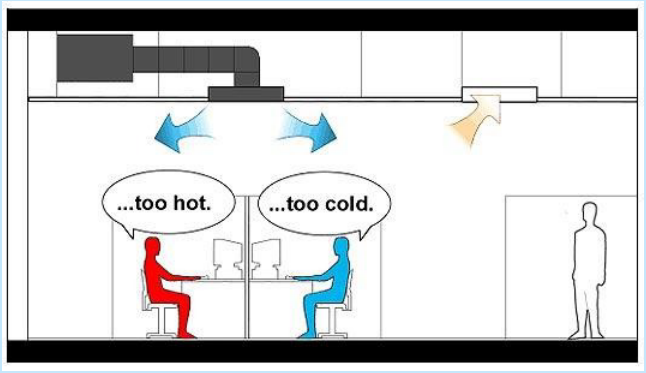
Lesson 9

DESIGN CONDITIONS

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INDOOR DESIGN CONDITION

- **Purpose**
  - Comfort Air-conditioning
  - Process (Industrial)
  - Refrigeration







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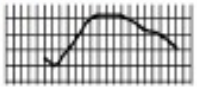
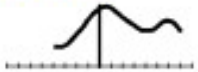
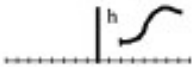
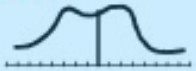
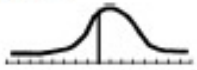
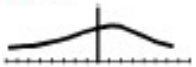
INDOOR DESIGN CONDITION

- Reference sources
  - CIBSE Guides
  - ASHRAE handbooks
  - Table 2.1 is an extract from the HVAC Applications Handbook

TABLE 2.1 – COMMERCIAL & PUBLIC BUILDINGS

General Category	Specific Category	Inside Design Conditions		Air Movement	Circulation, air changes per hour
		Winter	Summer		
Dining and Entertainment Centers	Cafeterias and Luncheonettes	21 to 23°C 20 to 30% rh	26°C[d] 50% rh	0.25 m/s at 1.8 m above floor	12 to 15
	Restaurants	21 to 23°C 20 to 30% rh	23 to 26°C 55 to 60% rh	0.13 to 0.15 m/s	8 to 12
	Bars	21 to 23°C 20 to 30% rh	23 to 26°C 50 to 60% rh	0.15 m/s at 1.8 m above floor	15 to 20
	Nightclubs and Casinos	21 to 23°C 20 to 30% rh	23 to 26°C 50 to 60% rh	below 0.13 m/s at 1.5 m above floor	20 to 30
	Kitchens	21 to 23°C	29 to 31°C	0.15 to 0.25 m/s	12 to 15[8]
Office Buildings		21 to 23°C 20 to 30% rh	23 to 26°C 50 to 60% rh	0.13 to 0.23 m/s 4 to 10 L/(s·m^2)	4 to 10
Museums, Galleries, Libraries and Archives	Average	20 to 22°C 40 to 55% rh		below 0.13 m/s	8 to 12
	Archival	See Chapter 21		below 0.13 m/s	8 to 12

TABLE 2.1 – COMMERCIAL & PUBLIC BUILDINGS

Noise[e]	Filtering Efficiencies (ASHRAE *Standard* 52.1)	Load Profile	Comments
NC 40 to 50[e]	35% or better	Peak at 1 to 2 PM 	Prevent draft discomfort for patrons waiting in serving lines
NC 35 to 40	35% or better	Peak at 1 to 2 PM 	
NC 35 to 50	Use charcoal for odor control with manual purge control for 100% outside air to exhaust =35% prefilters	Peak at 5 to 7 PM 	
NC 35 to 45[f]	Use charcoal for odor control with manual purge control for 100% outside air to exhaust =35% prefilters	Nightclubs peak at 8 PM to 2 AM Casinos peak at 4 PM to 2 AM Equipment, 24 h/day	Provide good air movement but prevent cold draft discomfort for patrons
NC 40 to 50	10 to 15% or better		Negative air pressure required for odor control (also see Chapter 31)
NC 30 to 45	35 to 60% or better	Peak at 4 PM 	
NC 35 to 40	35 to 60% or better	Peak at 3 PM 	
NC 35	35% prefilters plus charcoal filters 85 to 95% final[i]	Peak at 3 PM 	

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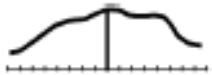
TABLE 2.1 – OFFICE BUILDINGS

indoor Design Condition (Summer)	23 - 26°C, 50 – 60% rh
indoor Design Condition (Winter)	21 - 23°C, 20 – 30% rh
Air Movement	0.13 – 0.23 m/s
Air Movement	4 – 10 l/s/m^2
Circulation, ACH	4 – 10
Noise	NC 30 - 45
Filtering Efficiencies	35 – 60% or better
Load Profile	Peak around 4 pm

TABLE 2.1 – COMMERCIAL & PUBLIC BUILDINGS

Bowling Centers		21 to 23°C 20 to 30% rh	24 to 26°C 50 to 55% rh	0.25 m/s at 1.8 m above floor	10 to 15
Communication Centers	Telephone Terminal Rooms	22 to 26°C 40 to 50% rh	22 to 26°C 40 to 50% rh	0.13 to 0.15 m/s	8 to 20
	Radio and Television Studios	21 to 23°C 40 to 50% rh	23 to 26°C 45 to 55% rh	0.13 to 0.15 m/s	15 to 40
Transportation Centers (also see Chapter 13)	Airport Terminals	23 to 26°C 30 to 40% rh	23 to 26°C 40 to 55% rh	below 0.13 m/s at 3.7 m above floor	8 to 12
	Ship Docks	21 to 23°C 20 to 30% rh	23 to 26°C 50 to 60% rh	0.13 to 0.15 m/s at 1.8 m above floor	8 to 12
	Bus Terminals	21 to 23°C 20 to 30% rh	23 to 26°C 50 to 60% rh	0.13 to 0.15 m/s at 1.8 m above floor	8 to 12
	Garages	4 to 13°C	27 to 38°C	0.15 to 0.38 m/s	4 to 6 Refer to NFPA
Warehouses		Inside design temperatures for warehouses often depend on the materials stored.			1 to 4

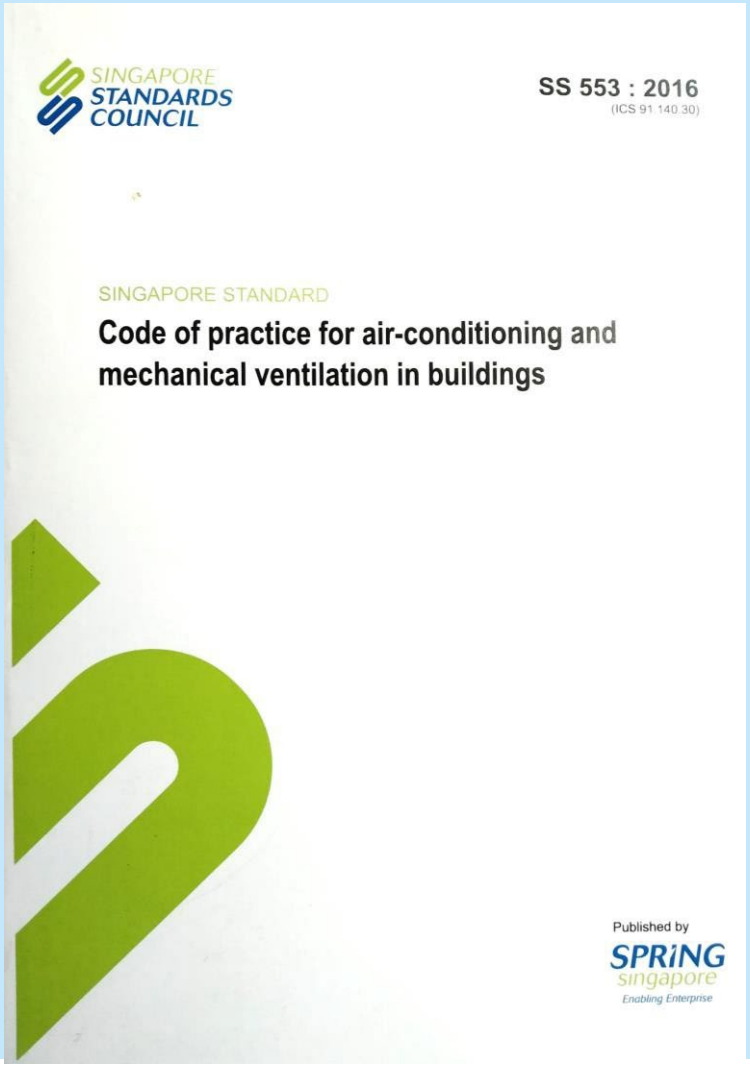
TABLE 2.1 – COMMERCIAL & PUBLIC BUILDINGS

NC 40 to 50	10 to 15%	Peak at 6 to 8 PM 	
to NC 60	85% or better	Varies with location and use	Constant temperature and humidity required
NC 15 to 25	35% or better	Varies widely because of changes in lighting and people	Constant temperature and humidity required
NC 35 to 50	35% or better and charcoal filters	Peak at 10 AM to 9 PM 	Positive air pressure required in terminal
NC 35 to 50	10 to 15%	Peak at 10 AM to 5 PM 	Positive air pressure required in waiting area
NC 35 to 50	35% with exfiltration	Peak at 10 AM to 5 PM 	Positive air pressure required in terminal
NC 35 to 50	10 to 15%	Peak at 10 AM to 5 PM 	Negative air pressure required to remove fumes; positive air in pressure adjacent occupied spaces
to NC 75	10 to 35%	Peak at 10 AM to 3 PM 	

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INDOOR DESIGN CONDITION

- SS 553 : 2016 - Code of Practice for Air-conditioning and Mechanical Ventilation in Buildings



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SS 553 : 2016

Sub-clause 7.1.3 of SS 553:2016 states:

- Normal design temperature - 23°C to 25°C
- Design RH - solar load assumed to be zero, while rest of the load components remain at their design values, resulting RH can be predicted ≤ 65%.
- Air speed for sedentary occupants ≤ 0.30 m/s.

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SS 553 : 2016

Clause 7.2 - in operation

- Dry bulb temperature - 23°C to 25°C
- Air movement ≤ 0.30 m/s at 1500 mm from floor
- Average RH ≤ 65% for new buildings and 70% for existing buildings.

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INDUSTRIAL (PROCESS AIR-CONDITIONING)

- Preferably obtain requirements from client.
- Handbooks e.g. Table 3.2 of ASHRAE handbook of HVAC Applications

TABLE 2.2

Table 1 Design Requirements for Industrial Air Conditioning[1]

Process	Dry Bulb, °C	rh, %
ABRASIVE		
Manufacture	26	50
CERAMICS		
Refractory	43 to 66	50 to 90
Molding room	27	60 to 70
Clay storage	16 to 27	35 to 65
Decalcomania production	24 to 27	48
Decorating room	24 to 27	48

Use high-efficiency (MERV 13 or better) in decorating room. To minimize the danger of silicosis in other areas, a dust-collecting system or medium-efficiency particulate air filtration may be required.

DISTILLING		
General manufacturing	16 to 24	45 to 60
Aging	18 to 22	50 to 60

Low humidity and dust control are important where grains are ground. Use high-efficiency filtration for all areas to prevent mold spore and bacteria growth. Use ultrahigh-efficiency filtration where bulk flash pasteurization is performed.

ELECTRICAL PRODUCTS		

Process	Dry Bulb, °C	rh, %
FOUNDRIES*		
Core making	16 to 21	
Mold making		
Bench work	16 to 21	
Floor work	13 to 18	
Pouring	4	
Shakeout	4 to 10	
Cleaning room	13 to 18	

*Winter dressing room temperatures. Spot coolers are sometimes used in larger installations.

In mold making, provide exhaust hoods at transfer points with wet-collector dust removal system. Use 280 to 380 L/s per hood, with a target capture velocity of approximately 2.5 m/s.

In shakeout room, provide exhaust hoods with wet-collector dust removal system. Exhaust 190 to 240 L/s in grate area. Room ventilators are generally not effective.

In cleaning room, provide exhaust hoods for grinders and cleaning equipment with dry cyclones or bag-type collectors. In core making, oven and adjacent cooling areas require fume exhaust hoods. Pouring rooms require two-speed powered roof ventilators. Design for minimum of 10 L/s of floor area at low speed. Shielding is required to control radiation from hot surfaces. Proper introduction of air minimizes preheat requirements.

TABLE 2.2

ELECTRICAL PRODUCTS		
Electronics and x-ray		
Coil and transformer winding	22	15
Semiconductor industry	21	45
Electrical instruments		
Manufacture and laboratory	21	50 to 55
Thermostat assembly and calibration	24	50 to 55
Humidistat assembly and calibration	24	50 to 55
Small mechanisms		
Close tolerance assembly	22*	40 to 45
Meter assembly and test	24	60 to 63
Switchgear		
Fuse and cutout assembly	23	50
Capacitor winding	23	50
Paper storage	23	50
Conductor wrapping with yarn	24	65 to 70
Lightning arrester assembly	20	20 to 40
Thermal circuit breakers assembly and test	24	30 to 60
High-voltage transformer repair	26	5
Water wheel generators		
Thrust runner lapping	21	30 to 50
Rectifiers		
Processing selenium and copper oxide plates	23	30 to 40

*Temperature to be held constant.

Dust control is essential in these processes. Minimum control requires medium-efficiency filters (MERV 11 or better). Degree of filtration depends on the type of function in the area. Smaller tolerances and miniature components suggest high-efficiency particulate air (HEPA) filters.

FLOOR COVERING	
Linoleum	
Mechanical oxidizing of linseed oil*	32 to 38
Printing	27
Stoving process	70 to 120

*Precise temperature control required.

Medium-efficiency particulate air filtration is recommended for the stoving process.

ing is required to control radiation from hot surfaces. Proper introduction of air minimizes preheat requirements.

FUR		
Drying	43	
Shock treatment	−8 to −7	
Storage	4 to 10	55 to 80

Shock treatment or eradication of any insect infestations requires lowering the temperature to −8 to −7°C for 3 to 4 days, then raising it to 16 to 21°C for 2 days, then lowering it again for 2 days and raising it to the storage temperature.

Furs remain pliable, oxidation is reduced, and color and luster are preserved when stored at 4 to 10°C.

Humidity control is required to prevent mold growth (which is prevalent with humidities above 80%) and hair splitting (which is common with humidities lower than 55%).

GUM		
Manufacturing	25	33
Rolling	20	63
Stripping	22	53
Breaking	23	47
Wrapping	23	58

LEATHER		
Drying	20 to 52	75
Storage, winter room temperature	10 to 16	40 to 60

After leather is moistened in preparation for rolling and stretching, it is placed in an atmosphere of room temperature and 95% relative humidity.

Leather is usually stored in warehouses without temperature and humidity control. However, it is necessary to keep humidity sufficiently low to prevent mildew. Medium-efficiency particulate air filtration is recommended for fine finish.

LENSES (OPTICAL)		
Fusing	24	45
Grinding	27	80

TABLE 2.2

Table 1 Design Requirements for Industrial Air Conditioning[1] (*Continued*)

Process	Dry Bulb, °C	rh, %
MATCHES		
Manufacture	22 to 23	50
Drying	21 to 24	60
Storage	16 to 17	50

Water evaporates with the setting of the glue. The amount of water evaporated is 8 to 9 kg per million matches. The actual match production rate must be known to determine the actual moisture load in the space.

PAINT APPLICATION		
Lacquers: Baking	150 to 180	
Oil paints: Paint spraying	16 to 32	80

The required air filtration efficiency depends on the painting process. On fine finishes, such as car bodies, high-efficiency particulate air filters are required for the outdoor air supply. Other products may require only low- or medium-efficiency filters.

Makeup air must be preheated. Spray booths must have 0.5 m/s face velocity if spraying is performed by humans; lower air quantities can be used if robots perform spraying. Ovens must have air exhausted to maintain fumes below explosive concentration. Equipment must be explosion-proof. Exhaust must be cleaned by filtration and solvents reclaimed or scrubbed.

PHOTO STUDIO		
Dressing room	22 to 23	40 to 50
Studio (camera room)	22 to 23	40 to 50
Film darkroom	21 to 22	45 to 55
Print darkroom	21 to 22	45 to 55
Drying room	32 to 38	35 to 45
Finishing room	22 to 24	40 to 55
Storage room (black and white film and paper)	22 to 24	40 to 60
Storage room (color film and paper)	40 to 50	40 to 50
Motion picture studio	22	40 to 55

The above data pertain to average conditions. In some color processes, elevated temperatures as high as 40°C are used, and a higher room temperature is required.

Conversely, ideal storage conditions for color materials necessitate refrigerated or deep-freeze temperatures to ensure quality and color balance when long storage times are anticipated.

Heat liberated during printing, enlarging, and drying processes is removed through an independent exhaust system, which also serves the lamp houses and dryer hoods. All areas except finished film storage require a minimum of medium-efficiency particulate air filters.

PLASTICS		
Manufacturing areas		
Thermosetting molding compounds	27	25 to 30
Cellophane wrapping	24 to 27	45 to 65

In manufacturing areas where plastic is exposed in the liquid state or molded, high-efficiency particulate air filters may be required. Dust collection and fume control are essential.

PLYWOOD		
Hot pressing (resin)	32	60
Cold pressing	32	15 to 25
RUBBER-DIPPED GOODS		
Manufacture	32	
Cementing	27	25 to 30*
Dipping surgical articles	24 to 27	25 to 30*
Storage prior to manufacture	16 to 24	40 to 50*
Testing laboratory	23	50*

*Dew point of air must be below evaporation temperature of solvent.

Solvents used in manufacturing processes are often explosive and toxic, requiring positive ventilation. Volume manufacturers usually install a solvent recovery system for area exhaust systems.

TEA		
Packaging	18	65

Ideal moisture content is 5 to 6% for quality and mass. Low-limit moisture content for quality is 4%.

TOBACCO		
Cigar and cigarette making	21 to 24	55 to 65*
Softening	32	85 to 88
Stemming and stripping	24 to 29	70 to 75
Packing and shipping	23 to 24	65
Filler tobacco casing and conditioning	24	75
Filter tobacco storage and preparation	25	70
Wrapper tobacco storage and conditioning	24	75

*Relative humidity fairly constant with range as set by cigarette machine.
Before stripping, tobacco undergoes a softening operation.

TOXIC AGENTS/HAZARDOUS MATERIALS		
Process areas	20 to 39	<65

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TABLE 2.2

Coil & Transformer Winding	22°C, 15% rh
Semiconductor Industry	21°C, 45 % rh
Motion Picture Studio	22°C, 40 – 55% rh
Plastics–Thermosetting compound	27°C, 25 – 30% rh
Plastics-Cellophane wrapping	24 - 27°C, 45 – 65% rh

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OUTDOOR DESIGN CONDITION

- Reference sources
  - ASHRAE handbooks
  - SS 553 : 2016 (Singapore Only)

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OUTDOOR DESIGN CONDITION – ASHRAE – TABLE 2.4

- Column for Cooling provides 0.4%, 1.0% and 2.0% dry bulb temps and mean co-incident wet-bulb temps. Represent conditions on hot and mostly sunny days. Used for cooling load calculations.

Station	Lat	Long	Elev	Heating DB		Cooling DB/MCWB					
						0.4%		1.0%		2.0%	
				99.6%	99%	DB	MC WB	DB	MC WB	DB	MC WB
Singapore											
Singapore Changi International	1.37N	103.98E	7	23.3	23.8	33.2	26.3	33.0	26.2	32.2	26.2

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OUTDOOR DESIGN CONDITION – ASHRAE – TABLE 2.4

- Values of dry bulb temp. corresponding to various annual percentiles represent value that is exceeded on average by indicated % of total number of hours in a yr (8760 hrs).

24h x 365days

- 0.4%, 1.0% and 2.0% values are exceeded on average 35, 88 and 175 hours.

0.02 x 8760

Station	Lat	Long	Elev	Heating DB		Cooling DB/MCWB					
						0.4%		1.0%		2.0%	
				99.6%	99%	DB	MC WB	DB	MC WB	DB	MC WB
Singapore											
Singapore Changi International	1.37N	103.98E	7	23.3	23.8	33.2	26.3	33.0	26.2	32.2	26.2

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OUTDOOR DESIGN CONDITION – ASHRAE – TABLE 2.4

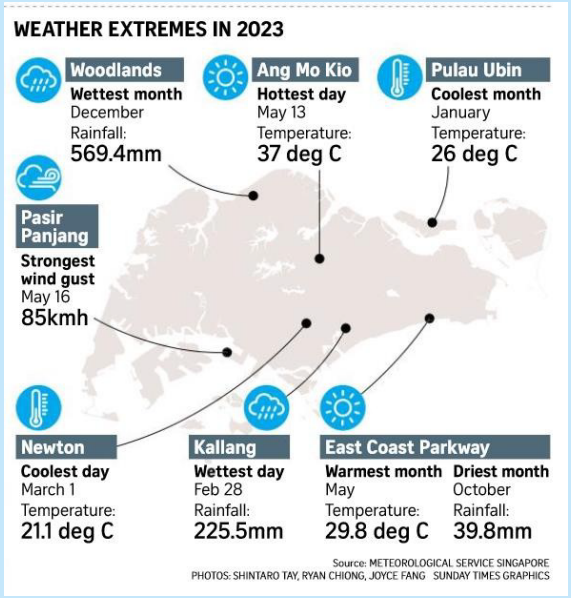
- If 2.0% data are used (ie design condition can cover 98% of time in a year), design dry bulb and wet bulb temperatures for Singapore are 32.2°C and 26.2°C respectively.

Station	Lat	Long	Elev	Heating DB		Cooling DB/MCWB					
						0.4%		1.0%		2.0%	
				99.6%	99%	DB	MC WB	DB	MC WB	DB	MC WB
Singapore											
Singapore Changi International	1.37N	103.98E	7	23.3	23.8	33.2	26.3	33.0	26.2	32.2	26.2

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OUTDOOR DESIGN CONDITION – SS 553 : 2016

- Design outdoor air conditions : 32°C db, 26°C wb with a daily range of 8°C.
- Dry-bulb temp. value is not exceeded more than 2% of total hours during months of June to September and wet-bulb value is average of coincident wet-bulb temperature occurring at design dry-bulb temperature.
- Dry bulb value is 0.2°C and wet bulb 0.2°C lower than ASHRAE handbook.



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OUTDOOR AIR TEMPERATURE – DAILY VARIATION

